

Homework Assignment 4

BAN 620 - DATA MINING

Group 2

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1. Develop various classification models using kNN and Trees to predict chattering.

Model 1: The client (osum) suggested to focus on the evaporator systems (evaporator 1,2,3,4) as it was the critical system in the plant.

- Approach: The accessing information about evaporator system
- Method:
 - I. The accessing information about evaporator system was available in the “data visualization” sheet of the given dataset.
 - II. Filtering all the records with (evaporator 1,2,3,4) from the “grouping of alarms” column from the “data visualization” sheet.
 - III. Getting the unique list of TagNames (110 in number) which are only having (evaporator 1,2,3,4) alarms from the “data visualization” sheet
 - IV. This unique list of TagNames are now matched with the training data. resulting in filtering of TagNames (“so” column in training data) which have evaporator 1,2,3,4 grouping of alarms (assumption). This results in 6054 rows of data.
 - V. evap_df with 6054 rows of data.
 - VI. Running knn and decision trees on evap_df .
 - VII. Predictors: 'atd', 'm', 'level', 'pressure', 'temperature', 'others', 'hour_7_12', 'hour_13_18', 'hour_19_24', '2nd_week', '3rd_week' '4th_week'
 - VIII. Target variable: 'chb'
 - IX. Decision tree 1(evap_df):
 - a. Accuracy on validation data: 0.8613
 - X. Decision tree 2(evap_df- grid search):
 - a. Accuracy on validation data: 0.879
 - XI. Knn:
 - a. K=3, Accuracy on validation data: 0.8980, False Negative Rate: 0.18
 - b. BEST K =6, Accuracy on validation data: 0.90132, False Negative Rate: 0.21

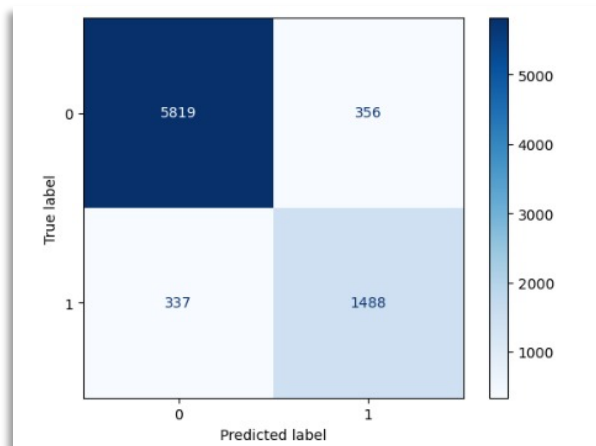
Model 2: Prioritizing Nametags most prone to chattering for better analysis of chattering behaviour.

- Approach: Ranking the alarm tags based on the absolute number of chattering alarms (chb = 1) to prioritize tags most prone to chattering
- Method:
 - I. Selecting the TOP 10 most frequent chattering alarm TagNames in the 'so' column of the training data.
 - II. These TOP 10 tagnames represent 58% of the chattering alarms in the training data.
 - III. Renaming all other TagNames other than TOP 10 most frequent chattering alarm TagNames as other_alarms. This enables to focus of chattering behaviour better.
 - IV. Dataframe: osum_df , 20000 rows , 60:40 train validation split
 - V. Predictors: 'atd', 'm', 'level', 'pressure', 'temperature', 'others', 'hour_7_12', 'hour_13_18', 'hour_19_24', '2nd_week', '3rd_week', '4th_week', 'so_FAL-3020', 'so_FAL-3452A', 'so_FALL-3452A', 'so_LDAH-3708', 'so_PAH-3200', 'so_PAL-3356', 'so_PAL-3390', 'so_PDAH-3190', 'so_PDIT-3091', 'so_XASS-3140'
 - VI. Target variable: 'chb'
 - VII. Decision tree (osum_df normal)
 - a) Accuracy on validation data: 0.904
 - VIII. Decision tree (osum_df grid search)
 - a) Accuracy on validation data: 0.896
 - IX. Knn:
 - a) k=3, Accuracy on validation data: 0.904, FNR: 0.21
 - b) best K =4, Accuracy on validation data: 0.905, FNR: 0.29

Model 3:

- Approach: applying category simplification
- Method:
 - I. Simplifying a categorical variable by grouping similar values into broader categories. For example, in "so" columns in values like 'XI-3057' and 'XI-3058' were reduced to just 'XI' by removing the numeric suffixes.
 - II. By Simplifying the column name we are reducing the number of unique "so" values from 452 to 52.
 - III. Renaming all other TagNames other than TOP 10 most frequent chattering alarm TagNames as other_alarms. This enables to focus of chattering behaviour better.
 - IV. Dataframe: o_osum_df , 20000 rows , 60:40 train validation split

- V. Predictors: 'atd', 'm', 'level', 'pressure', 'temperature', 'others','hour_7_12', 'hour_13_18', 'hour_19_24', '2nd_week', '3rd_week','4th_week', 'so_AAHH', 'so_AAL', 'so_AT', 'so_DAH', 'so_DAHH', 'so_EAL','so_ESDV', 'so_FAH', 'so_FAL', 'so_FALL', 'so_FIC', 'so_FIT', 'so_FSL','so_FT', 'so_HTCP', 'so_HTP', 'so_IT', 'so_KM', 'so_LAH', 'so_LAHH','so_LAL', 'so_LALL', 'so_LALLL', 'so_LDAH', 'so_LIT', 'so_LSHH','so_LT', 'so_MM', 'so_PAH', 'so_PAHH', 'so_PAL', 'so_PALL', 'so_PDAH', 'so_PDIT', 'so_PIT', 'so_PM', 'so_PT', 'so_SC', 'so_ST', 'so_TAH','so_TAHH', 'so_TAL', 'so_TDAH', 'so_TIT', 'so_TT', 'so_UA', 'so_UXA','so_VAH', 'so_VAHH', 'so_VIT', 'so_XA', 'so_XASS', 'so_XI', 'so_XV','so_ZA'
- VI. Target variable: 'chb'
- VII. Decision tree (o_osum_df normal)
 - a) Accuracy on validation data: 0.888
- VIII. Decision tree (o_osum_df grid search)
 - a) Accuracy on validation data: 0.903
- IX. Random forest (o_osum_df)
 - a) Accuracy on validation data: 0.915
- X. Gradient boosting (o_osum_df)
 - a) Accuracy on validation data: 0.88
- XI. Knn:
 - a) k=3, Accuracy on validation data: 0.91, FNR: 0.18
 - b) best K =3



2. Logit model with all the predictors:

The below is the confusion matrix for the model developed with all the predictors using logistic regression which has an accuracy of 0.90

Confusion Matrix (Accuracy 0.9030)		
Actual	Prediction	
	0	1
0	5690	462
1	314	1534

6TH QUESTION NEW DATA

3. Comparing predictive accuracy:

Comparing all the models, we see that the accuracy of KNN with K=3 has the best accuracy with 0.91.

However,

Higher false negatives are more costly and risky i.e. a critical alarm going unnoticed amidst the chattering could lead to equipment damage, safety hazards, or even a complete plant shutdown. This would leave operators unprepared for the ensuing alarm flood, leading to distractions, delayed responses, and potentially missed opportunities to prevent disruptions. Therefore, it is more critical to minimize false negatives (missed alarms), even at the cost of allowing some false positives (unnecessary alarms).

Therefore, if we compare the false negative rates,

KNN with K=3 has 0.18 (337/337+1488)

Logit model with all predictors has 0.16 (314/314+1524).

Logit model with all predictors is better.

4. Predictors to use based on predictive accuracy.

Based on predictive accuracy and the analysis, the predictors that should be used are:

```
Index(['atd', 'chb', 'm', 'level', 'pressure', 'temperature', 'others',  
      'hour_7_12', 'hour_13_18', 'hour_19_24', '2nd_week', '3rd_week',  
      '4th_week', 'so_AAHH', 'so_AAL', 'so_AT', 'so_DAH', 'so_DAHH', 'so_EAL',  
      'so_ESDV', 'so_FAH', 'so_FAL', 'so_FALL', 'so_FIC', 'so_FIT', 'so_FSL',  
      'so_FT', 'so_HTCP', 'so_HTP', 'so_IT', 'so_KM', 'so_LAH', 'so_LAHH',  
      'so_LAL', 'so_LALL', 'so_LALLL', 'so_LDAH', 'so_LIT', 'so_LSHH',  
      'so_LT', 'so_MM', 'so_PAH', 'so_PAHH', 'so_PAL', 'so_PALL', 'so_PDAH',  
      'so_PDIT', 'so_PIT', 'so_PM', 'so_PT', 'so_SC', 'so_ST', 'so_TAH',  
      'so_TAHH', 'so_TAL', 'so_TDAH', 'so_TIT', 'so_TT', 'so_UA', 'so_UXA',  
      'so_VAH', 'so_VAHH', 'so_VIT', 'so_XA', 'so_XASS', 'so_XI', 'so_XV',  
      'so_ZA'])
```

These features are relevant and provide sufficient information for the Decision trees model to make accurate predictions.

we got the best accuracy of **0.9168**

Confusion Matrix (Accuracy 0.9168)

	Prediction	
Actual	0	1
0	5819	356
1	337	1488

5. Meaning of the logit coefficients for at least two coefficients

a. Variable: flow

Coefficient: -0.995912

Interpretation: For every one-unit increase in the flow variable, the log odds of the chattering being

$Y = 1$.

$Y = 1$ decrease by approximately 0.996 holding all other variables constant.

In terms of probability:

A negative coefficient implies that higher values of flow are associated with a lower probability of

$Y = 1$.

$Y = 1$ (in this case, belonging to the class labeled as 1).

The impact can also be calculated in terms of odds: the odds of $Y = 1$

$Y = 1$ are multiplied by $e^{-0.995912} \approx 0.37$ meaning the odds decrease by about 63%.

Odds Ratio:

One unit increase in flow is $e^{-0.995912} \approx 0.37$ which means, one unit increase in flow, the odds of belonging to chattering alarm decreases by a factor of 0.37.

b. **Variable:** temperature

Coefficient: 0.947673

Interpretation: For every one-unit increase in the temperature variable, the log odds of the chattering being $Y = 1$.

$Y = 1$ increase by approximately 0.948, holding all other variables constant.

Odd's ratio:

One unit increase in temperature is $e^{0.947673} \approx 2.58$ which means, one unit increase in temperature the odds of belonging to chattering alarm increases by a factor of 2.58.

6. Predicting odds for New Data:

The below is the new dataset of 5 rows that we have used to predict the odds.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
SO	ATD	CHB	M	Alarm Tag Type	Flow	Level	Pressure	Temperature	Others	H	Hour:0-6	Hour:7-12	Hour:13-18	Hour:19-24	Week	1st Week	2nd week	3rd week	4th week	
FAL-3110	1.00	0	6	Others	0	0	0	0	1	0-6	1	0	0	0	0 4th Week	0	0	0	1	
EAL-3000	10.00	1	10	Flow	1	0	0	0	0	13-18	0	0	1	0	0 2nd Week	0	1	0	0	
FIT-3480	1.00	0	7	Flow	1	0	0	0	0	13-18	0	0	1	0	0 1st Week	1	0	0	0	
Xi-3057	2.00	0	7	Flow	1	0	0	0	0	19-24	0	0	0	0	1 4th Week	0	0	0	1	
EAL-3000	1.00	0	12	Pressure	0	0	1	0	0	0-6	1	0	0	0	0 3rd Week	0	0	1	0	

Using the model we have fit above to make predictions on this new data, the below is the accuracy we have achieved = 0.80

	actual	p(0)	p(1)	predicted
0	0	0.962059	0.037941	0
1	1	0.939287	0.060713	0
2	0	0.968608	0.031392	0
3	0	0.955129	0.044871	0
4	0	0.756521	0.243479	0

```
] : #Use the classification summary function for training
classificationSummary(test_y, new_reg_pred)
```

Confusion Matrix (Accuracy 0.8000)

```

      Prediction
Actual 0 1
      0 4 0
      1 1 0

```

Calculating Odds:

	actual	p(0)	p(1)	odds	predicted
0	0	0.962059	0.037941	0.039437	0
1	1	0.939287	0.060713	0.064637	0
2	0	0.968608	0.031392	0.032409	0
3	0	0.955129	0.044871	0.046979	0
4	0	0.756521	0.243479	0.321840	0